

In 19th century, **George Cantor** was the first mathematician who gave the concept of set.



George Cantor

1.1 SET

Define set

We often use some words in our daily life that represent a collection of things like dinner set, sofa set, tea set, group of boys, a team of players, a bunch of keys, crowd of people, a cluster of trees, a flock of sheep. The words set, group, team, bunch, crowd, cluster, flock are used to denote collection of things in daily life.

In mathematics collection of things is called set.

In mathematics we describe a set as:

A set is a collection of **'well defined'** and **'distinct'** objects.

The term **'Well defined'** means a set must have some specific property so, that it can easily be identified whether or not an object belongs to the given set.

Whereas the word **'Distinct'** means different objects.



Let us discuss some examples of collection.

- (1) The students of class VI of your school. It is well defined because students of class VI of your school can only belong to it. Each student has unique identity. Thus it is a set.
- (2) The collection of favourite books. It is not well defined because a book may be favourite for one person but may not be for another. Thus this collection is not a set.
- (3) Name of days of week; Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday. It is a well defined sets because every day is the member of this set, which is distinct.

Recognize notation of a set and its objects/elements

English alphabets A, B, C, . . . , and Z are used to denote sets. The objects of the set are called elements or members of the set. The elements of the set are listed within brackets { } and separated by commas.

The symbol ' $\in$ ' is used to show the membership to a set and read as "belongs to" and ' $\notin$ ' is used for, "does not belong" to a set.

Let us consider the following examples.

Example 1:

- (i) $B = \{ 2, 4, 6, 8 \}$

The members of set B are 2, 4, 6 and 8.

As 2 belongs to set B

So, symbolically we write as $2 \in B$

As 3 does not belong to Set B.

So, symbolically we write as $3 \notin B$

Teacher's Note

Teacher should give some more examples of collections from class room and daily life to clarify the concept of set such as name of books in a bag for grade VI and name of instruments in geometry box..

(ii) $D = \{x, y, z\}$

The elements of set D are x , y and z .

Here, $x \in D$, but $p \notin D$

Describe tabular form of a set and demonstrate through examples

As we have already described that all the elements of a set are listed within $\{\}$ and separated by commas, this form of describing a set is called **tabular form**.

Let us demonstrate this form through examples.

Example 1: Set of the vowels of the English alphabet

In tabular form we write as $\{a, e, i, o, u\}$ or $\{i, a, u, e, o\}$

Example 2: Set of natural numbers upto 100

In tabular form we write as $\{1, 2, 3, \dots, 100\}$

Example 3: Set of names of three boys whose names start with "A".

In tabular form we write as $\{\text{Ali, Ahmed, Ahmer}\}$

Example 4: Set of name of days of a week.

In tabular form we write as

$\{\text{Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday}\}$

Example 5: Explain, why are the following not sets:

(1) $A = \{5, 6, 6, 7, 8\}$

(2) $B = \text{Collection of intelligent students}$

Solution:

(1) $A = \{5, 6, 6, 7, 8\}$

This is not set because the element 6 is **repeated**.

(2) $B = \text{Collection of intelligent students}$

This is not set because it can not be decided that which students are intelligent.

**Activity 1**For $A = \{2, 4, 6, 8, 10\}$ Fill in the blanks with symbols $\in$ or $\notin$.

- (i) $4 \in A$ (ii) $1 \quad A$ (iii) $6 \quad A$
 (iv) $9 \quad A$ (v) $2 \quad A$ (vi) $8 \quad A$

**Activity 2**

Write the given sets in tabular form.

Set	Tabular Form
(i) Set of first five English alphabets.	$\{a, b, c, d, e\}$
(ii) Set of all factors of 12	
(iii) Set of names of days of a week which start from "F"	
(iv) Set of first five natural numbers	

Points to remember

- The name of a set is denoted by capital letter of English alphabet.
- Every element of a set is separated by a comma ',' in tabular form.
- All the members of set are enclosed in brackets $\{ \}$ in tabular form.
- The order of writing elements does not matter in a set.
- Elements of set do not repeat itself.

Teacher's Note

Teacher should perform the given activities on blackboard with students and also give some more examples.

EXERCISE 1.1

1. Tick (✓) which of the following are sets?

- (i) Collection of the names of Presidents of Pakistan.
- (ii) Collection of names of captains of hockey teams of Pakistan.
- (iii) Collection of delicious dishes.
- (iv) Collection of intelligent students in your class.
- (v) Collection of greater numbers.
- (vi) Collection of English teachers in your school.

2. If $A = \{5, 6, 7, 8\}$ and $B = \{a, b, c, d\}$, fill in the blanks with the symbols $\in$ or $\notin$.

- (i) 5 _____ A
- (ii) 9 _____ A
- (iii) m _____ A
- (iv) c _____ B
- (v) b _____ B
- (vi) 7 _____ B

3. Write in tabular form.

- (i) Set of name of capitals of the provinces of Pakistan.
- (ii) Set of natural numbers from 50 to 70.
- (iii) Set of first ten natural numbers exactly divisible by 2.
- (iv) Set of letters in the word "pakistan".
- (v) Set of mathematics teachers in your school.
- (vi) Set of name of colours of national flag.

4. If $P = \{a, e, i, o, u\}$ and $Q = \{1, 2, 3, \dots, 10\}$ then which of the following statements are true or false.

- (i) $a \in P$ _____
- (ii) $1 \in Q$ _____
- (iii) $i \in Q$ _____
- (iv) $v \in P$ _____
- (v) $y \in Q$ _____
- (vi) $e \in P$ _____
- (vii) $3 \in P$ _____
- (viii) $7 \in Q$ _____

Define finite and infinite sets

Consider the sets given in the table 1.

Table 1

$A = \{2, 4, 6\}$,	$B = \{1, 2, 3, \dots\}$
$C = \{6, 4, 2\}$,	$D = \{x, y, z\}$
$E = \{1, 3, 5, 7, \dots\}$,	$F = \{5, 10, 15, 20, 25, \dots\}$

Let us consider set A. It has limited number of elements, so it is **finite set**.

A set which has limited number of elements is called finite set.

In table 1, C and D are also finite sets.

Now consider set $B = \{1, 2, 3, \dots\}$.

This set has unlimited number of elements. So it is **infinite set**.

A set which has unlimited number of elements is called infinite set.

In table 1, E and F are also infinite sets.

**Activity**

Tick (✓) the finite sets and cross (✗) the infinite sets.

- (i) Set of all citizen of Pakistan
- (ii) Set of all whole number less than 9999
- (iii) Set of drops of water in river
- (iv) The set of all English alphabets
- (v) $\{1, 2, 3, \dots\}$
- (vi) $\{d, o, r\}$
- (vii) $\{2, 4, 6, \dots\}$

Define empty/void/null sets and singleton**Table 2**

$A = \{ 5 \}$,	$B = \{ x \}$
C = Set of natural numbers less than 1	
D = Set of triangles with four sides	

In table 2, C has no element because there is no natural number less than 1.

So, C is **an empty set**, which is also called **null or void set**.

A set which has no element is called an empty or null or void set. It is denoted by $\emptyset$ or $\{ \}$.

In table 2, D is also empty set. Some other examples of empty set are:

Examples:

- (i) Set of squares with five sides.
- (ii) Set of letters of English alphabet before A.

In table 2, there is a set $A = \{ 5 \}$ which has only one element and is called **singleton set**.

A set having only one element is called singleton.

In table 2, set B is also singleton. Some other examples of singleton are:

Examples:

$$P = \{ 8 \} \quad Q = \{ a \}, \quad R = \{ 1 \} \quad \text{and} \quad S = \{ 0 \}$$

Define equal and equivalent sets

Table 3

$$A = \{1, 2, 3, 4\},$$

$$B = \{x, y, z\}$$

$$C = \{4, 3, 2, 1\},$$

$$D = \{a, e, i, o, u\}$$

$$E = \{2, 4, 6, 8, 10\},$$

$$F = \{y, z, x\}$$

Consider sets A and C from the table 3. All the elements of both sets are the same. So they are called **equal sets**.

Two sets A and B are called equal sets if they have all elements same. Symbolically we represent as $A = B$ and read as:
Set A is equal to set B

In table 3, B and F are also equal sets.

When we see D and E, they have same number of elements and these are called **equivalent sets**.

Two sets A and B are called equivalent sets if the number of elements of both sets are same. Symbolically we write as:
 $A \sim B$ and read as: Set A is equivalent sets to set B

Also, in the given table 3,

$$A \sim C \text{ and } B \sim F$$

All equal sets are also equivalent sets but all equivalent sets may not be equal sets.

For example in table 3, (i) $A = C$ and also $A \sim C$

(ii) $D \sim E$ but $D \neq E$.

Define subset and superset of a set

Consider the sets given in the following table 4.

Table 4

$$A = \{ 1 \} ,$$

$$B = \{ 1, 2, 3 \}$$

$$C = \{ 1, 2, 3, 4 \},$$

$$D = \{ 4, 3, 2, 1 \}$$

Let us see sets B and C. All the elements of B are also elements of C. So we say B is **subset** of C.

If each element of set A is also an element of set B then set A is called subset of set B. Symbolically we write as $A \subseteq B$.

In the given table 4,

$$A \subseteq B, B \subseteq C, C \subseteq D \text{ and } D \subseteq C$$

Remember that

Every set is subset of itself.

$$\text{As } B \subseteq B \text{ and } D \subseteq D$$

Again, if we consider B and D.

B is subset of D. So D is called **superset** of B.

If A is subset of B then the set B is called superset of A. Symbolically we write it as ' $B \supseteq A$ '

In the given table 4,

$$B \supseteq A \text{ and } D \supseteq B$$

Define proper and improper subsets of a set and demonstrate through examples.

Now we consider set B and set D in table 4

Here $B \subseteq D$ and $B \neq D$.

So set B is called proper subset of set D.

Proper Subset

If set A is subset of set B and also A is not equal to B then "A is proper subset of set B. Symbolically we write as $A \subset B$."

In the given table 4, $A \subset B$ and $B \subset D$

In table 4, we see that set D is subset of set C but set D is also equal to set C. So D is called improper subset of C.

Improper Subset

If set A is subset of set B and A is also equal to B then "A is called improper subset of set B".

In table 4, set C is improper subset of set D.



Activity

Identify and write the following sets, from the given table.

- | | |
|-------------------------|------------------------|
| (i) Two finite sets | (ii) Two infinite sets |
| (iii) Two empty sets | (iv) Two equal sets |
| (v) Two equivalent sets | (vi) Two singleton |
| (vii) Two subsets | (viii) Two super sets |

Set of natural numbers between 2 and 3, Set of factors of 1

Set of triangles with four sides, $D = \{2, 4, 6, 8\}$

$A = \{1, 3, 5, 7, \dots\}$, $B = \{6\}$, $E = \{5, 10, 15, 20, 25, \dots\}$

$C = \{a, b, c, d\}$, $F = \{4, 6, 2, 8\}$,

Set of first four letters of English alphabet

EXERCISE 1.2

1. Give reasons why following collections are not set.

- (i) $\{d, o, o, r\}$ (ii) $\{\star, O, \star, \Delta\}$
(iii) $\{a, f, d, a\}$ (iv) $\{2, 2, 3, 3, 4, 4\}$
(v) Set of beautiful birds (vi) Set of good players

2. Which of the following are finite or infinite sets?

- (i) $A = \{0, 1, 2, \dots, 50\}$.
(ii) $B = \{100, 200, 300, 400, \dots\}$.
(iii) Set of hair on the body of goat.
(iv) Set of legs of a cat.
(v) Set of stars in the sky.
(vi) Set of lines passing through a point.
(vii) Set of natural numbers greater than 20.
(viii) Set of all the cities of Pakistan.
(ix) Set of all the schools in Sindh.
(x) Set of all even numbers.

3. Which of the following are empty sets?

- (i) Set of students of your class over 20 years of age.
(ii) Set of letters after Z in English alphabet.
(iii) Set of children whose names start with "K" in your locality.
(iv) Set of name of solar calendar starting with Z.
(v) Set of fishes which live in sand.
(vi) Set of even numbers between 4 and 10.
(vii) $\{0\}$ (viii) $\{\emptyset\}$

4. Which pairs of the following are equal sets?

- (i) $\{1, 2, 3\}$ and $\{2, 3, 1\}$ (ii) $\{p, a, t\}$ and $\{t, a, p\}$
(iii) $\{k, i, t, e\}$ and $\{b, i, t, e\}$
(iv) $\{x, y, z\}$ and set of first three letters of English alphabet.
(v) Set of odd numbers less than 2 and $\{\}$

5. Which pairs of the following are equivalent sets?

- (i) $\{m, i, l, e\}$ and $\{n, i, l, e\}$ (ii) $\{1, 2, 3\}$ and $\{3, 2, 1\}$
- (iii) $\{\Delta, \square, \bigcirc, \star, \square\}$ and $\{1, 2, 3, 4, 5\}$
- (iv) $\{6, 66, 666, 6666\}$ and $\{666, 7777, 77\}$

6. Which of the following statements are true or false?

- (i) $B = \{m, o, n\}$ if B is Set of letters of word "MOON".
- (ii) $A \subseteq B$ if $A = \{1, 2\}$ and B is $\{2, 3, 4, 5\}$
- (iii) If $A = \{4, 5, 10\}$ and B is $\{5, 10, 20\}$ then $A \sim B$.
- (iv) If $A = \{0, 1, 2, 3, 5\}$ and B is Set of natural numbers less than 6, then $A = B$.
- (v) If X is equal to Y, then X is equivalent to Y.

7. Choose appropriate symbol to indicate the relation between sets ($=, \subseteq, \not\subseteq, \supseteq, \sim, \subset$)

- (i) $\{10, 20, 30\}$ _____ $\{5, 10, 15, 20, 25, 30\}$
- (ii) $\{7, 14, 21, 28, 35, 42, 49\}$ _____ Set of multiples of 7 less than 50.
- (iii) $\{5, 6, 7, 8, \dots\}$ _____ $\{7, 5, 8\}$
- (iv) $\{0\}$ _____ $\{\quad\}$
- (v) $\{t, e, a\}$ _____ $\{a, e, t\}$
- (vi) $\{\text{Hyderabad, Karachi, Sukkur}\}$ _____ Set of all cities of Sindh.
- (vii) $\{a, e, i, o, u\}$ _____ Set of letters of English alphabets.
- (viii) $\{11, 22, 33\}$ _____ $\{1, 2, 3, \dots\}$

REVIEW EXERCISE 1

1. Fill in the blanks.

- (i) If $A = \{a, b, c\}$ then a, b, c are _____ of a set A .
- (ii) A set which has no element is called _____ set.
- (iii) A set which has limited number of elements is called a _____ set.
- (iv) If two sets A and B have _____ elements, then they are said to be equal sets.

2. Choose the correct answer.

- (i) If B is set of vowels of English alphabet:
(a) $p \in B$ (b) $e \notin B$ (c) $b \in B$ (d) $a \in B$
- (ii) $\{f\}$ is called:
(a) null set (b) infinite set (c) subset (d) singleton set
- (iii) If sets A and B are equal, we use symbol
(a) $\in$ (b) $\subset$ (c) $\sim$ (d) $=$
- (iv) If $A = \{a, b, c\}$ and $B = \{a, b, c, d, e\}$ then
(a) $A = B$ (b) $A \sim B$ (c) $A \subset B$ (d) $B \subset A$

3. Write two examples of set.

4. Write the following sets in tabular form.

- (i) A is Set of all even numbers between 2 and 10.
- (ii) B is Set of all odd numbers from 1 to 17.
- (iii) C is Set of natural numbers divisible by 2 and less than 30.
- (iv) D is Set of months of a solar calendar beginning with letter "J".

5. Which of these sets are finite and which of these are infinite.
- (i) A is Set of members of your family
 - (ii) B is Set of all even numbers
 - (iii) C is Set of prime factors of 60
 - (iv) D is Set of all multiples of 7
 - (v) E is Set of one-digit natural numbers
6. Which of the following are empty sets.
- (i) L is Set of the names of days of a week which start with "S".
 - (ii) F is Name of months of a year having 32 days.
 - (iii) M is Set of odd numbers exactly divisible by 2.
 - (iv) N is Set of prime numbers exactly divisible by 2.
7. If $A = \{d\}$, $B = \{c, d\}$, $C = \{a, b, c\}$, $D = \{a, b\}$ then which of the following statements are true or false.
- (i) $D \subseteq C$ (ii) $A \not\subseteq C$ (iii) $B \subseteq D$ (iv) $C \not\subseteq D$
8. Indicate which of the following statements are true and false?
- (i) The members of any set are always of same kind.
 - (ii) The order of the elements of a set does not matter.
 - (iii) An object can be included in a set repeatedly.
 - (iv) Any collection of objects is called a set.
 - (v) The notation " $\in$ " is a symbol to indicate membership of a set.
 - (vi) A well-defined collection of distinct objects is called a set.
 - (vii) $\{a, b, c, d\}$ is not a set.
 - (viii) The notation " $\notin$ " is a symbol to indicate membership of a set.
 - (ix) 1, 2, 3, 4 is a set.

SUMMARY

- A set is a collection of well defined and distinct objects.
- Each object in a set is an element or member of the set.
- A set is described in two ways.
 - (i) Precisely described in words.
 - (ii) All the elements are listed down separately.
- Finite set has limited number of elements.
- Infinite set has unlimited number of elements.
- A set having no element is an Empty or Null set.
- Two sets are equal if they have all same elements.
- Two sets are equivalent if they have same number of elements.
- B is subset of A if every element of B is also an element of A.
- A set having only one element is called singleton set.

Symbols to Remember

$\in$	is member of
$\notin$	is not member of
$\emptyset$ or $\{ \}$	empty set or null set
$=$	is equal to
$\neq$	is not equal to
$\sim$	is equivalent to
$\subset$	is a subset of
$\subsetneq$	is a proper subset of
$\supset$	is a superset of
$\not\subset$	is not a subset of